

B2 - BACnet / Modbus (Advanced)

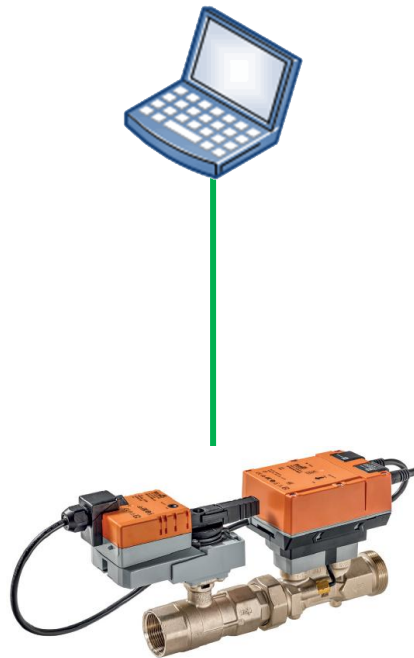
Modbus - Workshop I

2-way EPIV4

B2 - Modbus - Workshop I

2-way EPIV4 - Network

Modbus Poll



EP015R2+BAC

— Serial network (RS485)

B2 - Modbus – Workshop I

2-way EPIV4

1. Power your device.

 *See wiring diagrams in the datasheet*

2. Connect your device via Modbus RTU with your laptop.

 *Use RS485-USB adapter*

 *See wiring diagram in the datasheet*

3. Configure your actuator according to network overview.

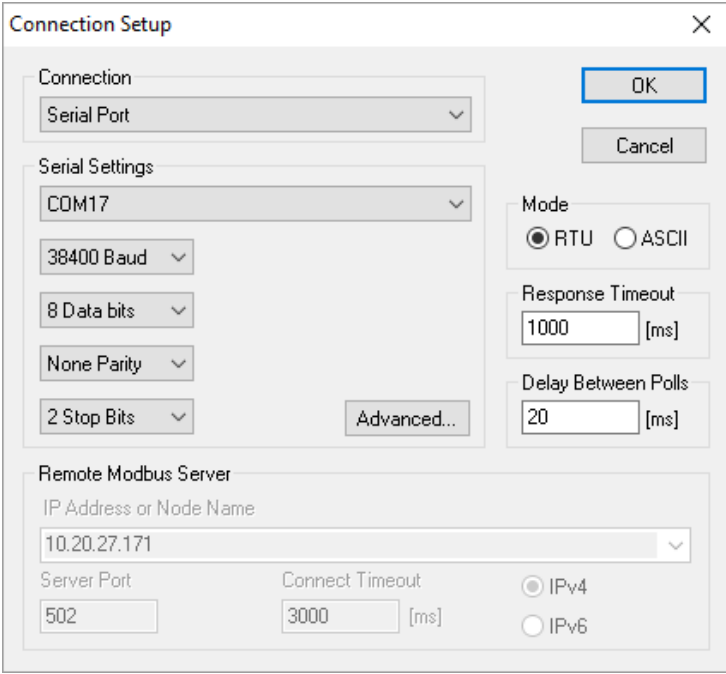
 *Find your appropriate configuration tool in the documents (LINK.10, Belimo Assistant 2, etc.)*

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2-way EPIV4

4. Start Modbus Poll. Open a *Serial Port* Connection.

- 💡 Check on which COM port your RS485-USB adapter is
- 💡 Pay attention to the Modbus settings on your actuator you just made



The image shows a 'Connection Setup' dialog box with the following settings:

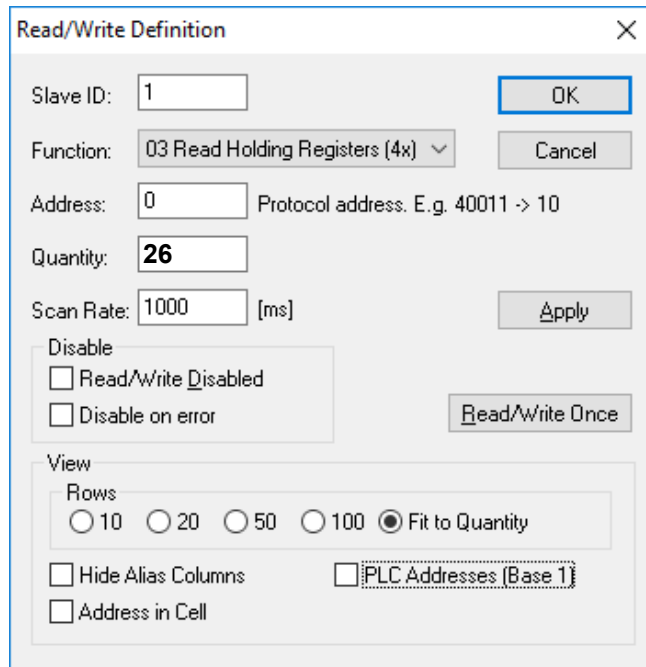
- Connection:** Serial Port
- Serial Settings:**
 - COM17
 - 38400 Baud
 - 8 Data bits
 - None Parity
 - 2 Stop Bits
 - Advanced...
- Mode:** RTU (selected), ASCII
- Response Timeout:** 1000 [ms]
- Delay Between Polls:** 20 [ms]
- Remote Modbus Server:**
 - IP Address or Node Name: 10.20.27.171
 - Server Port: 502
 - Connect Timeout: 3000 [ms]
 - IPv4 (selected), IPv6

Buttons: OK, Cancel

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5. Open a new window  and press the F8 key.
Make the settings according to the picture on the left.
Repeat the procedure with the settings as shown in the other pictures.



Read/Write Definition

Slave ID: 1 OK

Function: 03 Read Holding Registers (4x) Cancel

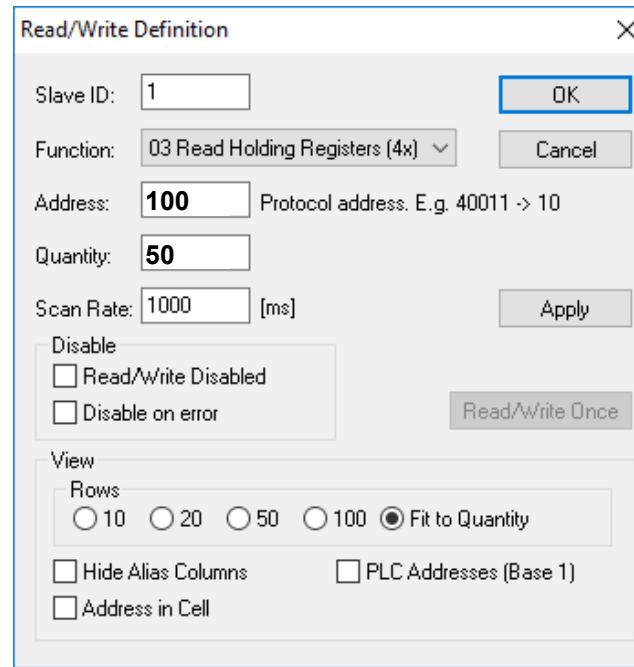
Address: 0 Protocol address. E.g. 40011 -> 10

Quantity: 26

Scan Rate: 1000 [ms] Apply

Disable
☐ Read/Write Disabled
☐ Disable on error Read/Write Once

View
Rows
☐ 10 ☐ 20 ☐ 50 ☐ 100 ☒ Fit to Quantity
☐ Hide Alias Columns ☐ PLC Addresses (Base 1)
☐ Address in Cell



Read/Write Definition

Slave ID: 1 OK

Function: 03 Read Holding Registers (4x) Cancel

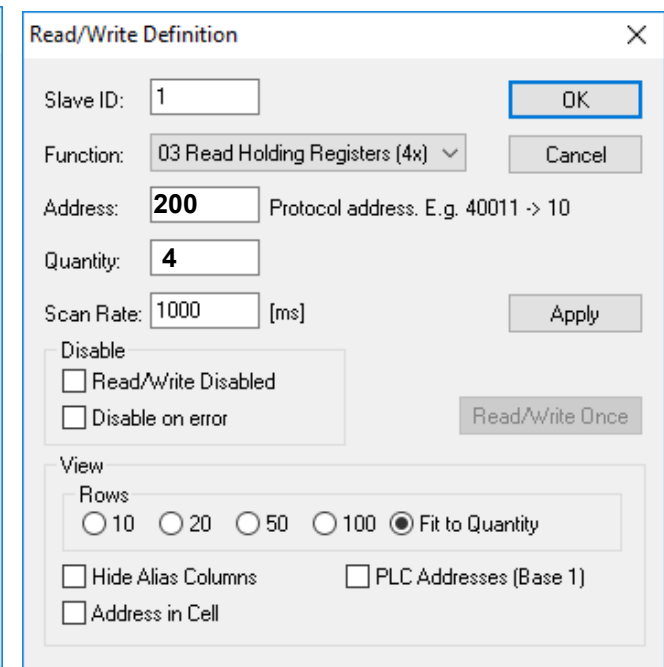
Address: 100 Protocol address. E.g. 40011 -> 10

Quantity: 50

Scan Rate: 1000 [ms] Apply

Disable
☐ Read/Write Disabled
☐ Disable on error Read/Write Once

View
Rows
☐ 10 ☐ 20 ☐ 50 ☐ 100 ☒ Fit to Quantity
☐ Hide Alias Columns ☐ PLC Addresses (Base 1)
☐ Address in Cell



Read/Write Definition

Slave ID: 1 OK

Function: 03 Read Holding Registers (4x) Cancel

Address: 200 Protocol address. E.g. 40011 -> 10

Quantity: 4

Scan Rate: 1000 [ms] Apply

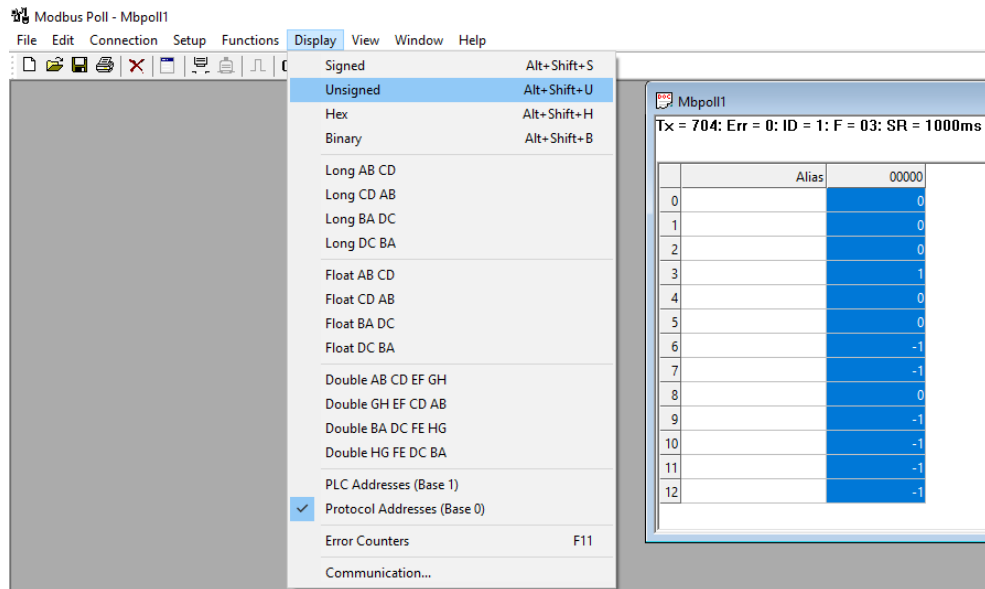
Disable
☐ Read/Write Disabled
☐ Disable on error Read/Write Once

View
Rows
☐ 10 ☐ 20 ☐ 50 ☐ 100 ☒ Fit to Quantity
☐ Hide Alias Columns ☐ PLC Addresses (Base 1)
☐ Address in Cell

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6. Display all values as *Unsigned Integer*.



7. What is the value in register No. 4

Value:

Interpretation:

8. Read the serial number (Register No: 101, 102 ,103)

Value:

Interpretation:

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9. Read Firmware Version.

Value:

10. Read Flow Body Temperature.

Value:

Interpretation:

11. Set [Communication watchdog] timeout to “30 seconds”.

Set [Bus Fail Action] to “Open”.

Observe the relative position of the actuator at Modbus Poll.

After opening completed, set [Bus override] to “Close”.

Which one has higher priority? Bus watchdog or Bus override?

Set back [Bus override] to “None”.

Set back [Bus watchdog] to “120s” as default.

Set back [Bus Fail Action] to “None”.
